Real-Time Video Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

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Agenda

01 Toward Real-Time H3 Generation
02 vLLM-Omni and ComfyUI Applications

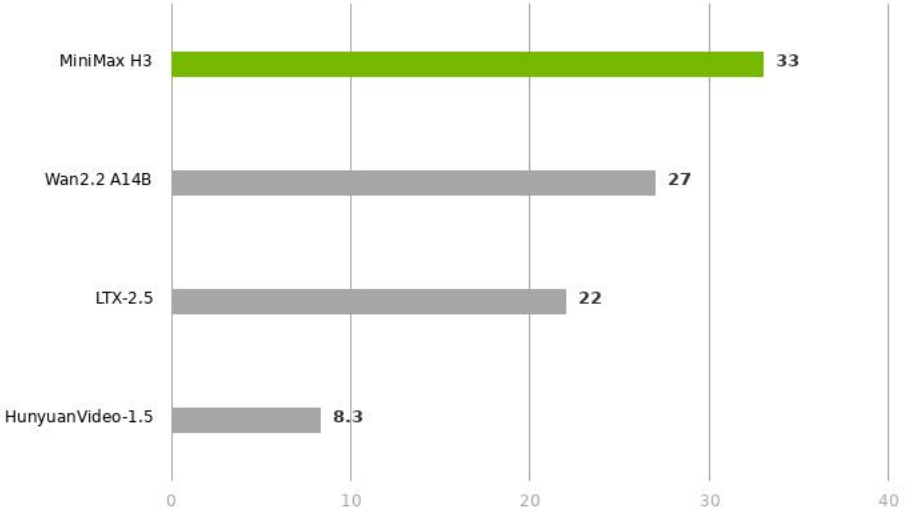


01 · Real-Time H3 Generation



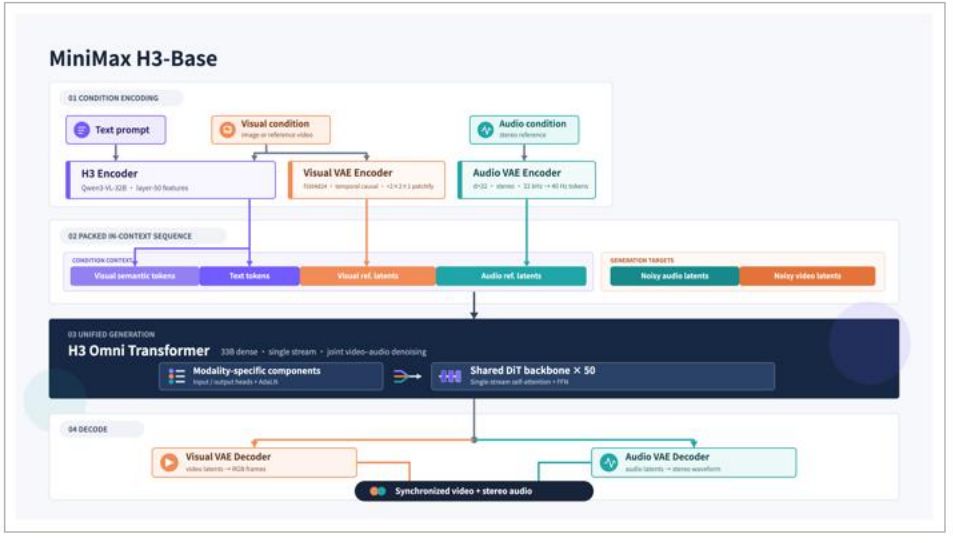
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

32B encoder → 33B dense Omni Transformer → visual/audio VAE. Fifty blocks × 50 steps keep compute and memory pressure high; about 13B AdaLN parameters can be cached

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: NIC → RDMA*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

Latency Breakdown

Communication and movement consume 61.7% of the mean GPU timeline

Stage	Mean Time per GPU*	Share
Mean GPU timeline*	248.3 s	100.0%
Repeated DiT loop	237.7 s	95.7%
Communication + movement	153.2 s	61.7%
DiT USP8 all-to-all (NCCL)	114.7 s	46.2%
AdaLN cache misses + layout	38.5 s	15.5%
Tensor compute	84.5 s	34.0%
Attention	45.3 s	18.3%
GEMM / MLP	39.2 s	15.8%
Encoders + VAE + other	10.6 s	4.3%

- Current trace uses a partial AdaLN cache; full schedule precompute requires a matched rerun
- Text, reference, and audio encoders are included in Encoders + VAE + other

Base Ref2VA · 960×544 · 124f @ 24 FPS · 50 steps · BF16 · 8× RTX PRO 5000 · local NCCL / RDMA off · *mean across 8 GPU timelines; NCCL includes sync / peer wait

PDD Removes 84% of Repeated DiT Work

H3 Ref2VA: 50 NFE → PDD: 8 NFE

Base Ref2VA

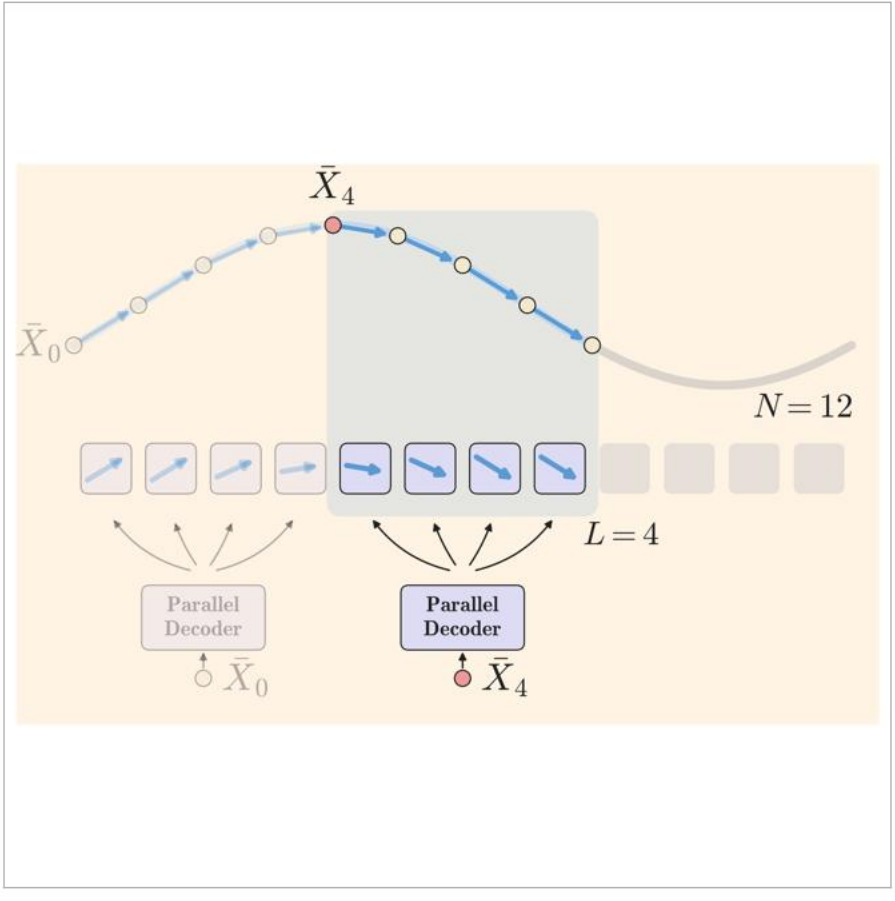
- **50 NFE × 50 blocks = 2,500**
  - Every attention, MLP and USP exchange repeats

Ref2VA PDD8

- **8 NFE × 50 blocks = 400**
  - N = 32, L = 4 → 8 evaluations

−84% repeated DiT block calls

- Matched latency, memory, and quality tests are pending



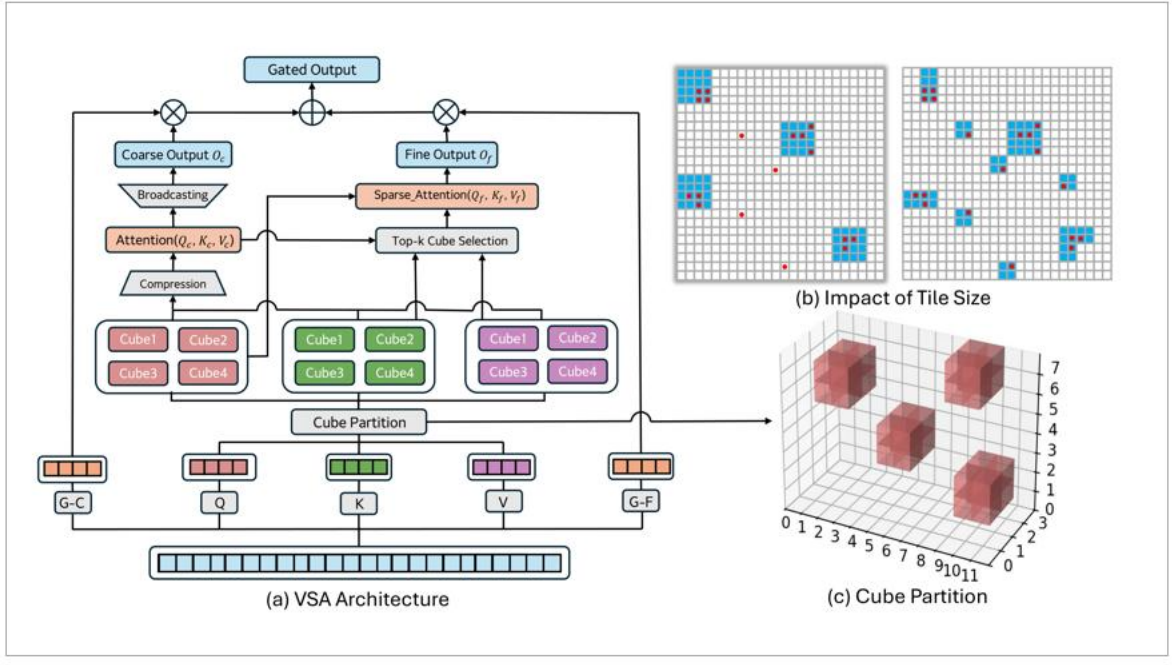
NVIDIA PDD · Parallel Decoder

Figure 2: N = 12 intervals, L = 4 per evaluation

Sources: NVIDIA PDD, Fig. 2 (arXiv:2607.26004) · Alibaba-PAI MiniMax-H3-Acc-LoRAs @ 335001f

After PDD, Attention Still Scores Every Pair

Video Sparse Attention (VSA) = trainable, content-aware block sparsity over 3-D video cubes



Dense Θ(S²) → VSA O((S/B)² + S·K·B)

At 544p, Svalid = 34,373, or 1.181B candidate pairs per head and block; USP8 still exchanges dense Q, K, V, O, and gate tensors

Source: Video Sparse Attention, arXiv:2505.13389, Fig. 1 · current H3 Ulysses communication path audited from source

QAD Quantizes Linears; Sage Quantizes VSA

Each path targets different SM120 kernels, and current evidence validates them separately

Path	Numerical Route	Effect	Validated Scope
QAD checkpoint	NVFP4 linears; BF16 exceptions	Recovers quantization error during distillation	8 forwards; PDD32; VSA 90%; tile 128
Sage VSA	Q/K INT8; V FP8; BF16 output	Quantizes tile64 fine attention after routing	SM120; d=128; opt-in; Ulysses
Precision island	Token refiner: BF16; VSA default: BF16	Keeps the sensitive short path in BF16	Same-seed quality check pending
Inter-GPU transport	BF16 Q/K/V before Sage; BF16 O after Sage	Keeps collective payloads in BF16	QAD and Sage do not reduce bytes

Sources: FastH3 Omni NVFP4-QAD rev 0087718 · NVIDIA ModelOpt QAD · vLLM-Omni SM120 Sage-VSA @ 63487a7

USP8 Issues Four All-to-Alls per DiT Block

Base50 runs 2,500 blocks at S = 34,432 on eight RTX PRO 5000 GPUs with local NCCL and no RDMA against a 5.167 s deadline

Scope	Tensor / Execution	A2A Rounds	Minimum Off-Rank Payload*	Rate at 5.167 s*
One DiT Block	Q / K / V forward + O reverse	4	1.728 GB	61.7 MB/rank/A2A
Base50	50 steps × 50 blocks = 2,500	10,000	4.319 TB	836.0 GB/s
PDD8	8 steps × 50 blocks = 400	1,600	691.1 GB	133.8 GB/s
Boundary	Current VSA / Sage keeps dense BF16 exchange	—	Logical bytes	Not achieved bandwidth

Source-derived @ vLLM-Omni H3 rev 63487a7f · logical one-way off-rank payload lower bound; not measured PCIe/NIC traffic or achieved bandwidth

RDMA Precheck Passes 96/96

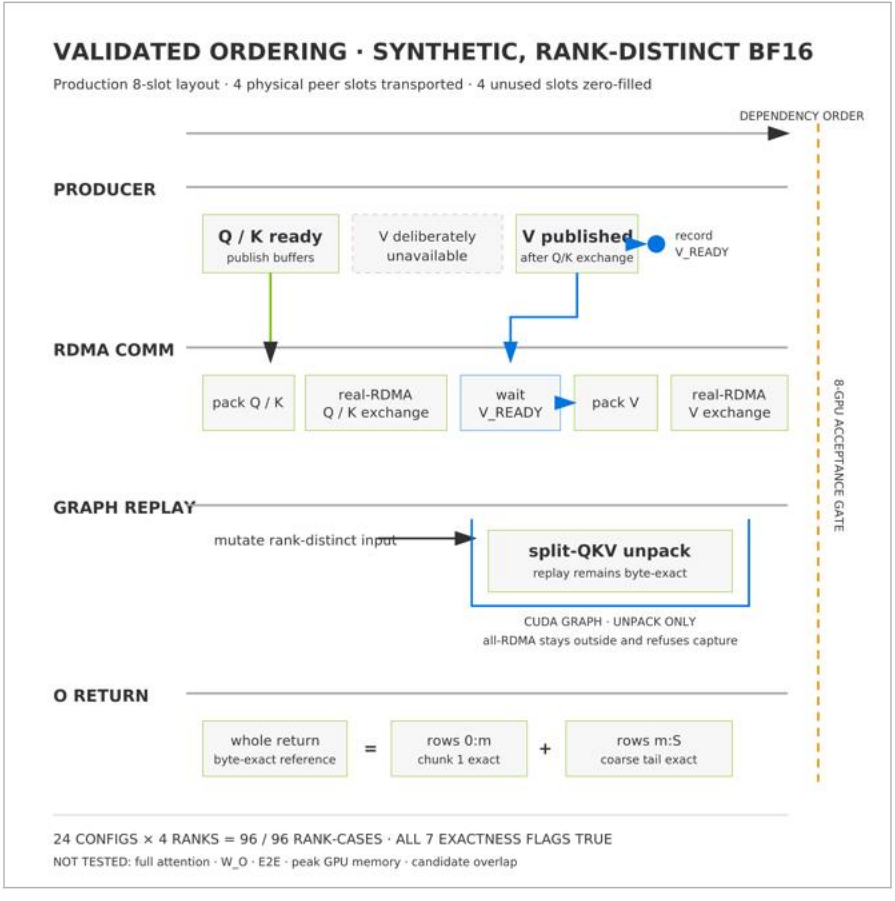
Four GPUs are byte-exact; eight-GPU H3 remains pending

Q/K first
- V waits for an explicit CUDA event

Graph replay
- Fixed-address unpack remains exact after the input changes

Tiled O return
- Whole and row-chunk returns match byte for byte

Eight-GPU acceptance gate
- Full attention + W_O, end-to-end latency, memory, and Nsight overlap



Four-GPU Real RDMA · 96 of 96 Rank Cases Exact

24 configs × 4 ranks · 8-GPU H3 candidate pending

Measured evidence: real-rdma-four-gpu-20261024-v2 · 2026-10-24 · candidate correctness precheck; latency and overlap not measured

Real-Time Demonstrated; Attribution Pending

Only Base50 uses the locked 960×544, 124-frame request; each optimized result uses a different workload

Status	Workload	System	Result / Limit
Measured baseline	960×544 · 124f · 5.167 s	Base50 · BF16 · USP8 · NCCL	247.956 s · RTF 47.992
15 s study	960×544 · 362f · 15.083 s	PDD8 · QAD · VSA · RDMA	14.974 s mean · RTF 0.993
10 s study	960×544 · 243f · 10.125 s	4-step Turbo · MXFP8 · Sage · RDMA	6.958 s · RTF 0.687
Portrait study	576×1024 · 119 new frames · 4.958 s	4-step MXFP8 · Sage · FP4 VAE · RDMA	4.459 s median · RTF 0.899
Required matched test	Lock image, audio, prompt, seed, shape, and request mode	PDD8 → VSA → QAD/Sage → RDMA	At least 3 runs per stage

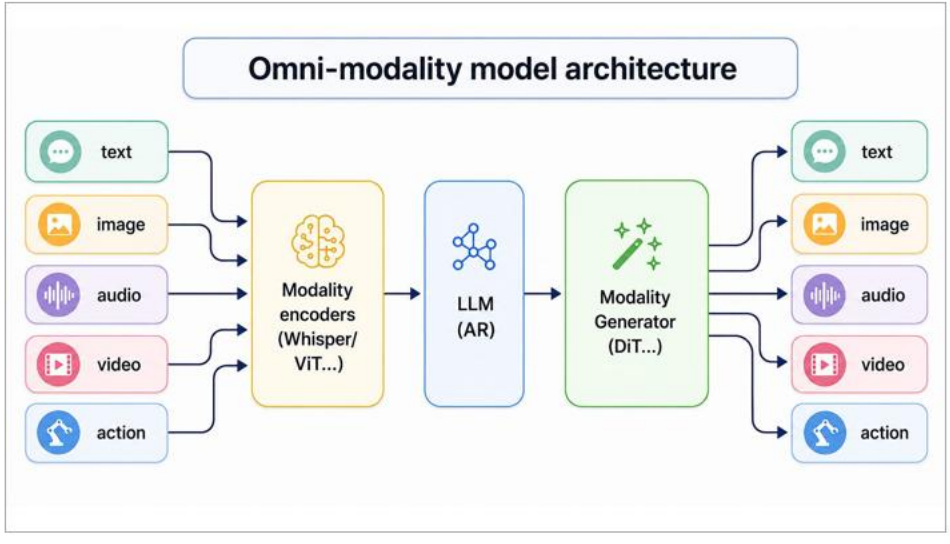
Sources: [A] Base50 r4 · [B] QAD 15 s cohort · [C] paired 10 s study · [D] portrait production · compare only within a row; timing and memory methods differ

02 · ComfyUI Apps With vLLM-Omni



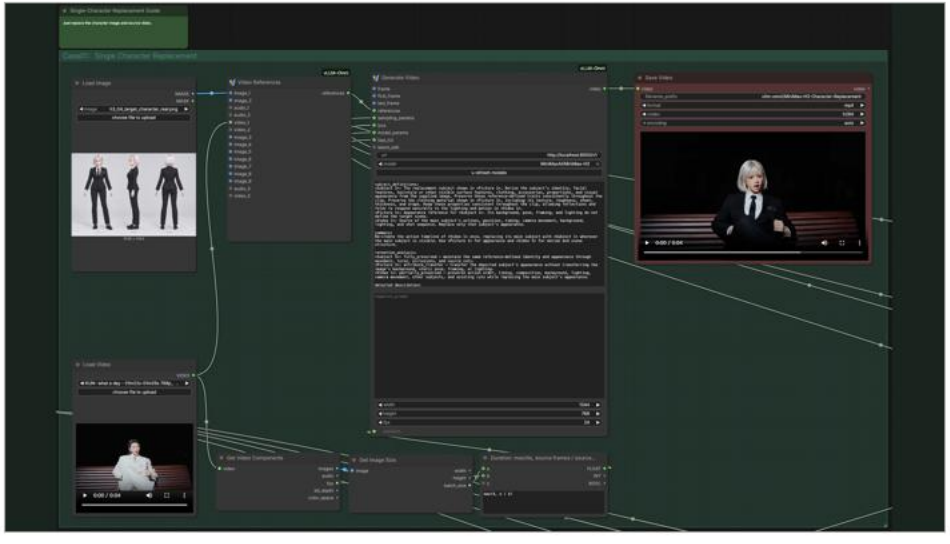
vLLM-Omni Brings Multi-GPU H3 to ComfyUI

Open PR #9194 adds character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI authors the workflow; vLLM-Omni runs scheduling, model execution, and multi-GPU deployment



ComfyUI Workflow · PR #9194

The workflow supports single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

